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**Seminar in Topics in Labor Economics and the Economics of
Education**

Did Remote Work Narrow the Motherhood Penalty?

**Evidence on the Intensive Margin from Israeli Labor Force
Survey Microdata, 2017–2023**

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Abstract

Harrington and Kahn (2025) ask whether the post-pandemic rise of work from home (WFH) reduced the motherhood penalty in the United States. This paper takes that question to Israel and to a different margin of labor supply. Using Central Bureau of Statistics (CBS) Labor Force Survey microdata on women aged 25–59 in 2017–2019 and 2021–2023, we estimate difference-in-differences (DiD) and triple-differences (DDD) models of employment and of usual weekly work hours for mothers relative to childless women, using the COVID-19-driven shift to remote work as the “post” period and a WFH-exposure index as the third difference (occupation-level for the hours model, demographic-cell-based for the employment model). On the extensive margin, the mother–non-mother employment gap did not change ($\text{Mother} \times \text{Post} = -0.0053$, SE 0.0052), and the employment DDD is underpowered rather than informative (minimum detectable effect about 26% of the baseline employment rate). On the intensive margin, employed mothers’ weekly hours rose by 0.83 hours relative to childless women (SE 0.18), and this increase is concentrated in WFH-exposed occupations: the triple interaction is 3.41 hours per unit of exposure (SE 0.93, $p < 0.001$), robust to a generalized trimming-bounds correction for selection into employment, with a 95% confidence interval for the identified set of $[1.50, 5.21]$. A gender placebo on fathers shows no such pattern. We read this as WFH access relaxing a binding time constraint for mothers who remain employed in teleworkable jobs.

1 Introduction

Harrington and Kahn (2025) pose a simple question for the United States: has the rise of work from home reduced the motherhood penalty in the labor market? Their answer is yes on the extensive margin and on income — a 10% rise in WFH raises mothers’ employment by roughly 0.78 percentage points relative to other women, concentrated in careers with high returns to hours and inflexible time demands, and employed mothers’ incomes rise by about 1.3% relative to other employed women. This paper asks the same question of a different country and a different margin. We use Israeli CBS Labor Force Survey microdata to test whether the post-COVID shift to remote work altered the motherhood penalty in Israel, and our headline result concerns *usual weekly work hours* among employed women rather than employment or income. It is therefore an extension of Harrington and Kahn’s design to another labor market and to the intensive margin, not a direct replication of their US finding: we also estimate the employment margin, but — for reasons of statistical power that we document rather than paper over — that specification is secondary.

The policy stakes are particularly high in Israel, which combines a highly productive, internationally integrated labor market with the OECD’s highest total fertility rate. A

labor market that induces mothers of young children to reduce employment or hours imposes a substantial aggregate cost. This cost is especially large in a country where a large and growing share of the working-age population is, at any given time, a parent of young children. Conversely, flexible work arrangements might measurably narrow the gap between mothers and other workers. If so, the finding bears directly on the design of remote-work policy, parental-leave rules, and public discourse on gender equality in the labor market. At the same time, Israeli institutional practice diverges sharply from US task-based predictions of which jobs can be done remotely — schools stayed in person by policy, for example — which matters for how “WFH exposure” should be measured and is one of the methodological contributions below.

Our research design exploits the pandemic as the moment at which remote work became a mainstream labor-market practice. Before 2020, working from home accounted for roughly 5% of workdays; after the pandemic it accounted for 25–30%. Unlike the acute, transitory disruption of 2020, WFH has proven to be a persistent feature of the post-pandemic labor market rather than a temporary accommodation (Barrero et al., 2021). This renders the question policy-relevant rather than merely historical. We therefore compare mothers and childless women across a pre-period of 2017–2019 and a post-period of 2021–2023, excluding 2020 as a transitional year of emergency, ad hoc arrangements that reflect improvisation rather than the settled remote-work practices whose effect we want to measure. (A second, practical reason is that no 2020 CBS extract exists for this project at all; Section 3 discusses what that implies for the exposure index.) The “post” dummy captures this structural shift toward remote work.

The core estimator is an individual-level DiD of the form $Y_{it} = \beta_0 + \beta_1 \text{Mother}_i + \beta_2 \text{Post}_t + \beta_3 (\text{Mother}_i \times \text{Post}_t) + X'_{it} \gamma + \varepsilon_{it}$, where β_3 is the differential post-shift change for mothers. To test whether remote work is the *mechanism*, we extend this to a triple-differences (DDD) model that interacts $\text{Mother} \times \text{Post}$ with an occupation’s WFH exposure, so that the parameter of interest is whether the narrowing of the penalty is larger in jobs that can actually be done from home.

Our findings are as follows. On the extensive margin there is nothing to see and, more importantly, nothing the design could have seen: the employment DiD is a precise zero, and the employment DDD has a minimum detectable effect roughly 8–10 times its point estimate. On the intensive margin, mothers’ hours rose by about 0.83 hours per week relative to childless women, but this average masks the real result: once occupational WFH exposure enters, the plain $\text{Mother} \times \text{Post}$ term collapses to essentially zero and the entire effect loads onto the triple interaction, 3.41 additional weekly hours per unit of exposure ($p < 0.001$). This estimate survives a selection-into-employment correction, two age-balance robustness specifications, and a gender placebo on fathers, whose corresponding coefficients are small and of the opposite sign (the fathers’ triple interaction is insignificant; their plain DiD is small, negative, and only marginally significant).

The remainder of the paper proceeds as follows. Section 2 places the question in the literature. Section 3 describes the data, the sample, and the construction of the WFH exposure measures. Section 4 sets out the DiD and DDD specifications and the selection correction. Section 5 reports the results, moving from the extensive to the intensive margin. Sections 6–7 interpret the findings and state their limitations, and Section 8 concludes.

2 Literature Review

The motherhood penalty. The child penalty is one of the most robust facts in labor economics. Kleven et al. (2019b) use Danish administrative data and an event-study design around the birth of the first child to show that women’s earnings fall by roughly 20% in the long run relative to men’s, with the gap fueled by reduced work hours, exit from the labor market, and moves into family-friendly, lower-paying jobs. Kleven et al. (2019a) extend the event-study approach across countries and find that the qualitative pattern is universal while its magnitude varies sharply across countries, and argue that differences in gender norms, rather than in family policy, are the more likely explanation. The Danish study matters here in particular because it identifies *hours* as one of the channels through which the penalty operates — which is why an intervention that changes the time cost of combining work and care could plausibly move the intensive margin even where it does not move employment.

Temporal flexibility as the mechanism. Goldin (2014) argues that the last chapter of gender convergence turns on the structure of jobs rather than on human capital: occupations that reward long, particular, and inflexible hours — where the marginal return to being physically present at a specific time is high — carry the largest gender gaps, and the gap would shrink if firms did not disproportionately reward temporal inflexibility. Remote work is, in this reading, precisely a change in temporal flexibility. It lowers commute friction, allows a virtual presence in demanding professions, and may reduce the stigma attached to a mother’s physical availability. It may also cut the other way, through lower visibility and slower advancement. Which effect dominates is an empirical question, and Goldin’s framework tells us where to look: in occupations where the work *can* be done remotely.

Does remote work narrow the gap in practice? The evidence on whether flexible-work arrangements realize this potential is mixed. In a field experiment on Italian white-collar workers, Angelici and Profeta (2024) find that randomly assigned flexible (“smart”) working arrangements shifted the division of household tasks. Women’s reported well-being also increased, suggestive of a genuine flexibility benefit. Real-time survey evidence

from the acute pandemic period tells a more ambiguous story. [Adams-Prassl et al. \(2020\)](#) show that WFH-feasible jobs were substantially better shielded from the 2020 employment and earnings shock. Women in WFH-feasible jobs, however, simultaneously absorbed a larger share of the additional childcare burden created by school and daycare closures. The same shock that protected employment could, in principle, degrade the quality of that employment for mothers specifically. [Alon et al. \(2020\)](#) formalize the macro logic for why COVID-19-type shocks hit mothers asymmetrically (sector composition plus school closures). They do not take a stand on whether WFH exposure specifically offsets or compounds this asymmetry. This paper is designed to adjudicate empirically between these two directions using post-acute-period (2021–2023) data, rather than the initial-shock period that most of this literature studies.

Measuring who can work from home. [Dingel and Neiman \(2020\)](#) classify occupations by whether their tasks can be performed entirely at home, using O*NET task content, and find that 37% of US jobs can be done from home. Their occupational classification, mapped to the two-digit ISCO-08 codes in the CBS data, is the starting point for our WFH-exposure index. As Section 3 explains, however, a US task-based measure misclassifies occupations where Israeli institutional practice diverged from what the tasks would predict, and we therefore calibrate it against realized Israeli behavior. For the Israeli labor market specifically, [Yaish et al. \(2021\)](#) document that the 2020 lockdown reallocated paid and unpaid labor within dual-earner households in gender-unequal ways. [Madhala and Bental \(2020\)](#) provide a descriptive occupation-level baseline of WFH feasibility among Israeli workers. [Buzaglo-Baris \(2023\)](#) shows that within-firm sorting, not occupation choice alone, is a first-order driver of the Israeli gender wage gap. This is a reminder that any occupation-based exposure measure captures only part of the relevant heterogeneity.

Remote work and the motherhood penalty. The paper closest to ours is [Harrington and Kahn \(2025\)](#), who show for the United States that the rise of WFH raised mothers’ employment relative to other women’s, that this is driven by fields with high returns to hours and inflexible demands on time, and that women who could work from home before childbirth are less likely to exit the workforce. We take their question to Israel. Our contribution is threefold: a different country with a distinctive combination of high fertility and high productivity; a different and, in the child-penalty literature, central margin — hours conditional on employment — with an explicit correction for the selection problem that conditioning on employment creates; and an exposure measure calibrated to Israeli practice rather than imported from US task content. Because the data span both pre- and multiple post-COVID years, they also allow separate identification of extensive- and intensive-margin responses — a distinction the WFH-and-gender literature above largely

does not draw.

Identification. The triple-differences estimator we use for the mechanism test is analyzed formally by [Olden and Møen \(2022\)](#), who clarify that the DDD does not require two parallel-trends assumptions but rather a single assumption that the *difference* between the two DiDs would have evolved in parallel absent treatment. Section 4 states the identifying assumptions in those terms.

3 Data and Institutional Context

3.1 Source and sample

The data are individual-level microdata from the Israeli Central Bureau of Statistics Labor Force Survey for the survey years 2017, 2018, 2019, 2021, 2022, and 2023. The analysis sample is women aged 25–59 (CBS age-group codes 3–7). A woman is coded as a *mother* if she has any child under 17 in the household, and as childless otherwise; the treatment group is mothers and the control group is childless women. *Post* equals one for survey years 2021 and later.

Two outcomes are studied. The primary outcome is *usual weekly work hours*, a continuous variable constructed from the CBS binned hours item by assigning each bin its median and imputing irregular-hours codes from observed medians within the same period; it is defined only for employed women. The secondary outcome is a binary employment indicator, equal to one if the respondent reports being employed and zero otherwise (unemployed and not-in-labor-force are both zero). Among employed women the pooled mean of weekly hours is 38.24 ($n = 288,400$) and the pooled employment rate is 0.7737 ($n = 372,741$).

Table 1 describes the sample on either side of the treatment/control split. Two features matter for what follows. Mothers are *more* likely to be employed than childless women (0.7828 versus 0.7574) but work *fewer* hours when they are (37.63 versus 39.38), so the penalty this paper studies is visible in the raw data on the intensive margin and runs the other way on the extensive one. And mothers are the better-educated group, with 47.97% holding an academic degree against 40.24% of childless women, which is why the controls matter: the unconditional hours gap understates the conditional one. The two groups are also imbalanced on age, mothers sitting about 0.44 of an age-group code younger, an imbalance Section 5 returns to because it tracks WFH exposure.

All regressions include the same set of categorical controls: marital status, religiosity, age group, district of residence, and highest educational attainment (six groups). Age enters as the categorical CBS age-group code because the extract contains no continuous age or birth-year variable. CBS survey weights are deliberately not applied in any outcome

regression; all reported estimates are unweighted estimates on the analysis sample rather than population-representative statistics (see Section 7).

Table 1: Summary statistics: women aged 25–59, by mother status

	Childless women	Mothers	Difference
<i>Panel A: outcomes</i>			
Observations	133,646	239,095	
Employment rate	0.7574	0.7828	0.0254
Usual weekly hours (employed), mean	39.38	37.63	−1.76
standard deviation	[12.87]	[12.77]	
<i>Panel B: characteristics</i>			
Age-group code (CBS 3–7), mean	5.40	4.96	−0.44
Arab (%)	17.73	17.78	0.05
<i>Education (%)</i>			
Below high school	11.37	6.63	−4.74
High school, no matriculation	12.70	11.95	−0.74
Matriculation (Bagrut)	22.73	18.82	−3.90
Post-secondary, non-academic	11.50	13.68	2.18
Academic degree (BA/MA/PhD)	40.24	47.97	7.73
Other / no certificate	1.47	0.95	−0.52

Israeli CBS Labor Force Survey, survey years 2017–2019 and 2021–2023, women aged 25–59. A woman is a mother if she has any child under 17 in the household. Weekly hours are conditional on employment; the standard deviation is in brackets. Education percentages sum to 100 within each column. The difference column is mothers minus childless women. All figures are *unweighted* counts on the analysis sample and are not population-representative (Section 7). Marital status, religion and district enter every regression as controls but are omitted here: the extract carries them only as unlabelled numeric CBS codes. Their full composition by mother status is reported in the replication output.

3.2 Why 2017–2019 versus 2021–2023

The window is designed around the pandemic’s role in making WFH a mainstream practice: working from home rose from about 5% of workdays before 2020 to 25–30% afterward. The post-period is therefore 2021–2023 and the pre-period 2017–2019. The year 2020 is excluded as a transitional year. There is also a data-driven reason: no 2020 CBS extract was acquired for this project, so 2020 could not be included even in principle. The consequence that matters for the design is discussed next.

3.3 Constructing WFH exposure

Since realized WFH status is itself a labor-market outcome — jointly determined with employment and hours — causal designs need a measure of WFH *exposure* that is not

chosen by the individual whose hours are being modeled. The mechanism test therefore requires an occupation-level measure of how exposed each job is to remote work. We build it in three steps, each producing a measure that is reported.

External. The [Dingel and Neiman \(2020\)](#) teleworkability score, via an O*NET/SOC-to-ISCO crosswalk, at the two-digit ISCO-08 level. This is pre-period by construction and immune to Israel’s own COVID-era behavior, but it is a US task-based measure and misclassifies occupations where Israeli institutional practice diverged: teaching (ISCO 23) scores 0.966 on the external index although Israeli schools stayed in person by Ministry of Education policy.

Calibrated (primary). We correct the external score only where the gap between it and the realized 2022–23 Israeli WFH share is large (greater than 0.5) *and* statistically distinguishable from sampling noise, using a one-sided cluster-robust test at the 95% level. Ten of the 40 two-digit occupations are swapped to their realized Israeli values. The corrections are concrete and checkable: teaching falls from 0.966 to 0.069, clerical support (ISCO 41) from 1.000 to 0.098, while ICT professionals (ISCO 25) go from 1.000 to 0.360. Because it draws on 2022–23 realized data, which sit inside the post-period, the calibrated measure is a documented compromise rather than a fully pre-treatment index; the exposure measures are built from a population that includes men as well as the women in the analysis sample, so that an occupation’s score is not constructed from exactly the people who enter the regression.

Realized. The realized Israeli WFH share by occupation, anchored to 2021 with a minimum of 200 observations per occupation. This is post-treatment by construction and is reported as a robustness check.

The original research plan specified anchoring the realized index to the 2020 WFH variable. Because no 2020 extract exists, the index is anchored to 2021 instead. This is a disclosed deviation with a stated justification — 2021 already flows through the validated pipeline, and a measure taken during 2020’s emergency disruption might reflect improvisation rather than each occupation’s settled remote-work potential — and a stated downside: by 2021 occupations had had a year to adapt, and if adaptation was uneven a 2021-based index could rank occupations differently than a 2020-based one would have. We return to this in [Section 7](#).

For the employment-outcome DDD, occupation is undefined for the non-employed, so an occupation-level regressor would condition the outcome on itself. That specification therefore uses a fourth measure: a pre-period (2017–2019) shift-share index by demographic cell (sex, age group, education, district, marital status, religiosity, and continent of birth — 8,884 cells), equal to the weighted mean of the calibrated occupation score over the cell’s pre-period occupational composition. It is defined for employed and non-employed women alike. The two DDDs thus use *different* exposure measures, and we are careful never to compare their coefficients as if they were on a common scale.

4 Empirical Strategy

4.1 Hypotheses

Two hypotheses follow directly from the flexibility framework in [Goldin \(2014\)](#) and the mixed empirical record above. **H1 (extensive margin):** occupational WFH exposure raises mothers' employment probability relative to other workers after 2021, if flexibility relaxes a binding participation constraint. **H2 (intensive margin):** even absent a detectable extensive-margin effect, WFH exposure raises mothers' hours conditional on employment, if the participation margin is structurally protected (e.g., by a furlough regime) while hours remain the flexible margin of adjustment.

4.2 Difference-in-differences

For outcome Y_{it} (employment, or weekly hours conditional on employment) we estimate

$$Y_{it} = \beta_0 + \beta_1 \text{Mother}_i + \beta_2 \text{Post}_t + \beta_3 (\text{Mother}_i \times \text{Post}_t) + X'_{it} \gamma + \varepsilon_{it}, \quad (1)$$

where X_{it} is the control vector described above and standard errors are clustered by individual identifier. β_3 is the DiD estimator: the differential change in the outcome for mothers relative to childless women after the shift to remote work. For the hours specification a positive β_3 indicates a narrowing of the penalty on the intensive margin; for employment it indicates a narrowing on the extensive margin. The employment DiD is also estimated separately for Jewish and Arab women.

4.3 Triple differences

To test whether the narrowing operates through remote work, we interact the DiD with occupational WFH exposure:

$$Y_{it} = \alpha + \text{Mother}_i \times \text{Post}_t \times \text{WFH_Exposure}_{j(i)} \text{ (all lower-order terms)} + X'_{it} \gamma + \varepsilon_{it}. \quad (2)$$

The coefficient on the triple interaction asks whether the post-shift change for mothers relative to non-mothers is larger in more exposed occupations. Following [Olden and Møen \(2022\)](#), the identifying assumption is that the mother/non-mother gap in high- versus low-exposure jobs would have evolved in parallel absent the shift to remote work. For the hours DDD, WFH_Exposure_j is the calibrated occupation-level score of [Section 3.3](#), the sample is employed women matched to an occupation, and standard errors are clustered by two-digit occupation — the level at which the regressor varies, so that individual-level clustering would understate the standard error. For the employment DDD, WFH_Exposure is the cell-based shift-share index, estimated on the full sample

of employed and non-employed women, with a Mother $\times$ age group interaction added to correct a documented pre-period age imbalance between mothers and non-mothers, in two variants: additive controls (Spec 1) and fully interacted age-by-education-by-district cell fixed effects (Spec 2). Standard errors are clustered on the age-by-education-by-district cell (about 210 clusters). We also report the exposure-specific minimum detectable effect, $\text{MDE} = \text{SE} \times (z_{0.975} + z_{0.80}) = 2.8016 \times \text{SE}$, so that a null coefficient can be read as “genuinely null” or “underpowered.”

4.4 Selection into employment: Lee bounds

Hours are observed only for the employed, and employment is itself an outcome of the same treatment. If remote work differentially pulls marginal mothers into employment after 2021, the post-period sample of employed mothers is compositionally different from the pre-period one for reasons unrelated to hours choices, biasing β_3 in the hours equation in a direction that is not signable a priori. We address this with trimming bounds in the spirit of Lee (2009), adapted to the two-by-two design. Let $s_{ab} = \Pr(\text{Employed} = 1 \mid \text{Mother} = a, \text{Post} = b)$. The parallel-trends-in-selection counterfactual for post-period mothers is $s_{11}^* = s_{10} + (s_{01} - s_{00})$. If the observed s_{11} exceeds s_{11}^* , the excess share $p = 1 - s_{11}^*/s_{11}$ is trimmed from the (Mother = 1, Post = 1) cell’s hours distribution — from the top to obtain a lower bound on β_3 , from the bottom to obtain an upper bound — and equation (1) is re-estimated on each trimmed sample. For the DDD we generalize this construction: the selection counterfactual is computed separately within each quartile of the cell-based exposure index (which, unlike the occupation-level index, is observable for the non-employed), each quartile’s post-period mothers are trimmed by that quartile’s own excess share, the trimmed subsets are recombined, and equation (2) is refit with the occupation-level regressor. Two different exposure measures thus serve two different roles. Because the parameter is only partially identified, we report, alongside each bound’s own cluster-robust confidence interval, the Imbens and Manski (2004) confidence interval for the identified set, which accounts jointly for the sampling uncertainty of both endpoints. The construction handles only *excess* selection in the treated cell, and assumes monotone selection; both limitations are stated in Section 7.

4.5 Diagnostics and placebo

We estimate event-study versions of equation (1) with Mother $\times$ year interactions (reference year 2019) for both outcomes and test the pre-period coefficients jointly. We re-estimate the hours DDD with an added Mother $\times$ age-group interaction and with pre-period age-group raking weights, in response to the age imbalance noted above; with the external and realized exposure indices in place of the calibrated one; and, as a gender placebo, on the male subsample with “mother” read as “father,” comparing the male and

female coefficients with two-sample z -tests (valid without a covariance correction because the samples are disjoint).

5 Results

Throughout, significance is denoted *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, · $p < 0.1$. Standard errors are in parentheses.

5.1 Extensive margin: the employment DiD

Table 2 reports equation (1) with employment as the outcome, for all women and separately for Jewish and Arab women. The DiD coefficient is -0.0053 (SE 0.0052) in the pooled sample: the mother/non-mother employment gap did not change after the shift to remote work. The estimate is precise — its standard error is about half a percentage point — so this is a tight null on the extensive margin, not an uninformative one. The stratified estimates are -0.0041 (SE 0.0058) for Jewish women and -0.0208 (SE 0.0138) for Arab women, neither significant.¹ Employment rose in the post-period for all women (Post = 0.0153^{***} pooled), and markedly so for Arab women (0.0601^{***}).

Table 2: Extensive-margin DiD: employment

	All women	Jewish women	Arab women
Mother $\times$ Post	-0.0053 (0.0052)	-0.0041 (0.0058)	-0.0208 (0.0138)
Mother	-0.0034 (0.0041)	0.0054 (0.0045)	0.0200 (0.0110)
Post	0.0153^{***} (0.0043)	0.0070 (0.0049)	0.0601^{***} (0.0111)
Constant	0.4926^{***} (0.0085)	0.5920^{***} (0.0119)	0.6300^{***} (0.0582)
Observations	364,784	283,975	65,598
R^2	0.22689	0.04336	0.24314

Dependent variable: employment indicator. Controls: marital status, religiosity, age group, district, education. Standard errors clustered by individual. Jewish/Arab samples defined by the CBS population-group code.

¹The religiosity control (**Dat**) is not identified within the Jewish-only subsample and enters with fewer categories in the Arab-only subsample, so the effective control set differs slightly across the stratified models.

5.2 Extensive margin: the employment DDD is underpowered

Table 3 reports the employment DDD with the cell-based exposure index. The triple interaction is -0.0257 (SE 0.0725) with additive controls and -0.0194 (SE 0.0722) with interacted cell fixed effects — small, negative, and nowhere near significant. The question is whether this is evidence of no mechanism or evidence of nothing. The minimum detectable effect answers it: at $\alpha = 0.05$ and 80% power, the design could detect a triple interaction no smaller than 0.2031 (Spec 1) or 0.2022 (Spec 2), which is about 26% of the baseline employment rate of 0.7737 and roughly 8–10 times the point estimates. Moving a mother from the least to the most exposed cell would have to change her relative post-2021 employment probability by a quarter of the baseline employment rate before this design could see it. That is not a plausible effect size for a labor-supply mechanism, so the null here is *uninformative*: the employment DDD cannot be read as evidence for or against a WFH-mediated employment effect. The $\text{Post} \times \text{WFH_Exposure}$ main effect, by contrast, is negative and significant in both specifications (-0.1555^{**} and -0.1516^{**}). This indicates a general post-2021 employment decline in high-exposure cells that is unrelated to motherhood specifically.

This is not for want of a working exposure measure. The cell-based index strongly predicts realized post-period WFH: a first-stage regression of the reference-week WFH indicator on exposure in the post-period yields 0.6557^{***} (SE 0.0954, $n = 110,051$), stable across 2021–2023 (interactions with 2022 and 2023 of 0.0118 and 0.0379, the latter marginal). Nor is it a specification failure: the exposure cells are deliberately finer than the fixed-effect cells, so the regressor retains within-cell variation (R^2 of exposure on the fixed-effect variables 0.285, VIF 1.40). An earlier version of the design, in which exposure was built on exactly the regression’s own control variables, had an MDE of 51% of baseline; successive refinements of the cell definition halved it to the current 26%, and cell-level aggregation was confirmed to add no power. The remaining limitation is structural: exposure assigned at the demographic-cell level simply does not vary enough. This is what motivates the intensive-margin analysis, where occupation-level exposure is available.

5.3 Intensive margin: hours

Pre-trends. Before turning to the estimates, Table 4 and Figure 1 report the hours event study. Relative to 2019, the $\text{Mother} \times \text{year}$ coefficient is -1.524^{***} in 2017 but a flat 0.1222 in 2018; a joint Wald test on the two pre-period coefficients rejects parallel trends over the full window ($F(2, 65,088) = 23.7$, $p = 5.2 \times 10^{-11}$). The shorter 2018–2019 window is consistent with parallel trends, and we state the identifying assumption on that basis. The same test on the employment outcome passes ($F(2, 79,069) = 0.63$, $p = 0.53$): the pre-trend problem is specific to hours. In the post-period the effect builds gradually — 0.3069 in 2021, 0.1080 in 2022, and 0.7435^{**} in 2023. Figure 1 plots the

Table 3: Extensive-margin DDD: employment $\times$ cell-based WFH exposure

	Spec 1: additive controls	Spec 2: interacted cell FE
Mother $\times$ Post $\times$ WFH_Exposure	−0.0257 (0.0725)	−0.0194 (0.0722)
Mother $\times$ Post	0.0065 (0.0168)	0.0044 (0.0166)
Mother $\times$ WFH_Exposure	−0.1404 (0.0803)	−0.0818 (0.0702)
Post $\times$ WFH_Exposure	−0.1555** (0.0566)	−0.1516** (0.0554)
WFH_Exposure	0.0131 (0.0565)	0.0056 (0.0490)
MDE ($\alpha = 0.05$, power 0.80)	0.2031	0.2022
MDE / baseline employment rate	26.3%	26.1%
MDE / point estimate	7.9 $\times$	10.4 $\times$
Observations	355,984	355,984
R^2	0.22103	0.22952

Dependent variable: employment indicator; full sample of employed and non-employed women. Both specifications include Mother $\times$ age-group interactions and the standard controls (Spec 2 absorbs age group, education, and district into interacted cell fixed effects). Standard errors clustered on the age-by-education-by-district cell. Baseline employment rate 0.7737.

same coefficients with their confidence intervals, where both the isolated 2017 dip and the gradual post-period build are visible directly.

The hours DiD. Column (1) of Table 5 reports equation (1) for weekly hours among employed women. Mothers work 2.223 fewer hours per week than childless women (Mother = −2.223***), the intensive-margin penalty itself. After the shift to remote work this gap narrowed: Mother $\times$ Post = 0.8261*** (SE 0.1830), so mothers’ weekly hours rose by about 0.83 hours more than non-mothers’ ($N = 281,750$). The raw cell means tell the same story — mothers’ hours rose from 36.85 to 38.52 while childless women’s rose from 39.01 to 39.80.

This estimate is, however, fragile to selection into employment. The employment rate of post-period mothers (0.7961) exceeds the parallel-trends-in-selection counterfactual ($s_{11}^* = 0.7717 + (0.7639 - 0.7518) = 0.7838$), so there is excess selection to trim — about 1.5% of the cell.² Table 6 reports the resulting Lee bounds: the lower bound is 0.2133 (SE 0.1808, 95% CI [−0.1411, 0.5678]) and the upper bound 1.3132 (SE 0.1814). The Imbens–Manski 95% confidence interval for the identified set is [−0.084, 1.612], which crosses zero. The plain hours DiD therefore does *not* survive a selection-on-employment correction, and we treat it as suggestive rather than as the substantive finding.

²Derived from the selection rates reported in the note to Table 6 ($1 - 0.7838/0.7961$), not directly exported by the pipeline.

Table 4: Event study: Mother $\times$ year, weekly hours (reference 2019)

Year	Coefficient	SE
2017	-1.524^{***}	0.2766
2018	0.1222	0.2140
2021	0.3069	0.2567
2022	0.1080	0.2620
2023	0.7435^{**}	0.2609
Joint test, 2017–2018 $F(2, 65,088) = 23.7, p = 5.2 \times 10^{-11}$		
Observations		281,750

Employed women only; standard controls; standard errors clustered by individual.

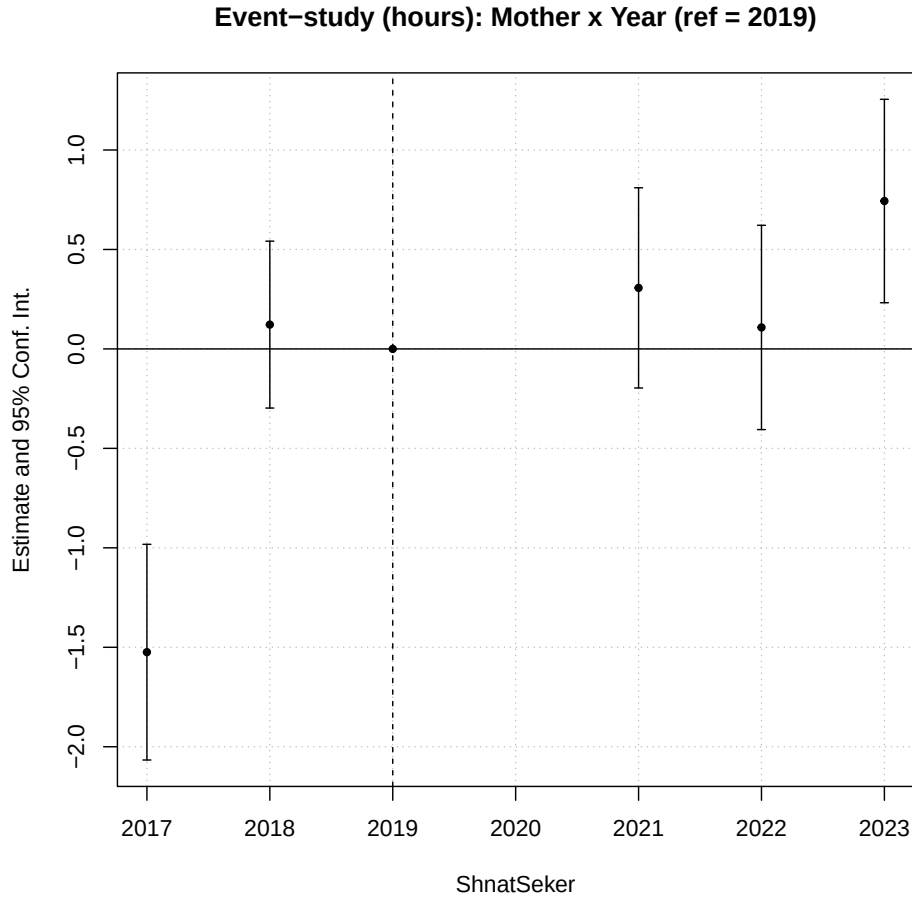


Figure 1: Event study: Mother $\times$ year interactions for weekly hours, reference year 2019

Notes: Employed women only; standard controls; standard errors clustered by individual. Vertical bars are 95% confidence intervals. The gap between 2019 and 2021 is the excluded 2020 survey year. Point estimates are those of Table 4.

Table 5: Intensive-margin DiD and DDD: weekly hours, employed women

	(1) DiD	(2) DDD, calibrated exposure
Mother $\times$ Post $\times$ WFH_Exposure		3.407*** (0.9302)
Mother $\times$ Post	0.8261*** (0.1830)	0.0153 (0.2893)
Mother $\times$ WFH_Exposure		−1.429 (1.143)
Post $\times$ WFH_Exposure		−0.8668 (0.8280)
WFH_Exposure		1.831 (4.153)
Mother	−2.223*** (0.1485)	−1.897*** (0.3650)
Post	0.7289*** (0.1503)	0.9153** (0.2746)
Constant	30.90*** (0.3198)	30.51*** (1.570)
MDE for triple interaction		2.6059
Observations	281,750	275,708
R^2	0.03836	0.03906
Clustering	individual	2-digit occupation

Dependent variable: usual weekly work hours, employed women only. Standard controls in both columns. Column (2) uses the calibrated occupation-level exposure index; 281,622 of 288,400 employed rows (97.6%) matched an occupation, 275,708 after listwise deletion on controls. MDE at $\alpha = 0.05$, power 0.80.

Table 6: Lee bounds on the hours DiD (Mother $\times$ Post)

Bound	Coefficient	SE	95% CI
Lower	0.2133	0.1808	[−0.1411, 0.5678]
Point (untrimmed)	0.8261	0.1830	[0.4675, 1.1846]
Upper	1.3132	0.1814	[0.9576, 1.6688]
Imbens–Manski 95% CI			[−0.084, 1.612]

Selection rates s_{ab} : $s_{00} = 0.7518$ ($n = 71,532$), $s_{01} = 0.7639$ (62,114), $s_{10} = 0.7717$ (129,856), $s_{11} = 0.7961$ (109,239).

The hours DDD: the paper’s headline result. Column (2) of Table 5 interacts the DiD with the calibrated occupation-level exposure index. The triple interaction is 3.407*** (SE 0.9302, $N = 275,708$): among employed mothers, those in more WFH-exposed occupations saw a larger post-2021 increase in weekly hours relative to childless women in the same occupations — about 3.4 additional hours per unit of exposure, where exposure runs from zero to one. Two features of the column are as informative as the headline. First, the bare Mother $\times$ Post term collapses from 0.826 to 0.0153 and becomes insignificant once the exposure interactions enter: the average hours effect in the DiD is not a broad, occupation-independent shift but is absorbed entirely into exposure heterogeneity. Second, the estimate is well inside the design’s power: the minimum detectable effect is 2.6059 hours (6.8% of mean weekly hours, or 76.5% of the point estimate), so unlike the employment DDD this design could and did detect an effect of the size observed.

Lee bounds for the DDD. Table 7 applies the generalized selection correction. Excess selection into employment among post-period mothers is present in all four quartiles of the cell-based exposure index but modest everywhere: the trim proportion ranges from 0.25% (quartile 2) to 4.61% (quartile 1, the least teleworkable jobs), and a total of 1,068 of 83,336 post-period employed mothers are trimmed. The bounds barely move: the lower bound is 3.4414 (SE 0.9927), the upper bound 3.4570 (SE 0.8975), and the Imbens–Manski 95% confidence interval for the identified set is [1.5035, 5.2091], which excludes zero comfortably. Unlike the plain DiD, the triple interaction survives the selection correction. This is the strongest single piece of evidence in the paper: the result is not an artifact of differential selection into employment.

Robustness. Table 8 collects the remaining checks. With the raw [Dingel and Neiman \(2020\)](#) external index in place of the calibrated one, the triple interaction is 0.9404 (SE 0.8649), positive but insignificant; with the realized 2021-anchored index it is 6.416* (SE 2.454). The direction is consistent across all three measures but the magnitude and significance are not, and we describe the result as directionally rather than point-estimate robust to the exposure measure. The calibrated measure is the one purpose-built to correct the external index’s known misclassification of Israeli occupations, and we lead with it. Adding a Mother $\times$ age-group interaction leaves the estimate essentially unchanged (3.367**, SE 0.9553); reweighting the sample to balance pre-period age groups between mothers and non-mothers attenuates it by about 17% to 2.836** (SE 0.9055), still significant at the 1% level. The age imbalance being corrected is real and tracks exposure — the pre-period mother/non-mother gap in the age-group code is -0.974 in the lowest exposure quartile ($t = -94.0$), -0.735 in the second, -0.169 in the third, and $+0.057$ in the highest — so the reweighted attenuation is worth reporting, but it does not overturn the result. An incidental finding of the age-interacted specification is a significant, non-

Table 7: Generalized Lee bounds on the hours DDD (Mother $\times$ Post $\times$ WFH_Exposure)

<i>Panel A: selection rates and trimming by cell-exposure quartile</i>							
Quartile	s_{00}	s_{01}	s_{10}	s_{11}	s_{11}^*	$n(\text{Mother}=1, \text{Post}=1)$	Trim
1	0.6284	0.6419	0.6045	0.6478	0.6180	21,655	4.61%
2	0.7624	0.7867	0.7773	0.8037	0.8016	28,540	0.25%
3	0.8167	0.8099	0.8651	0.8619	0.8584	34,293	0.41%
4	0.8674	0.8514	0.8538	0.8501	0.8377	21,815	1.45%
<i>Panel B: bounds on the triple interaction</i>							
Bound	Coefficient		SE		95% CI		
Lower	3.4414		0.9927		[1.4958, 5.3870]		
Point (untrimmed)	3.4077		0.9564		[1.5331, 5.2823]		
Upper	3.4570		0.8975		[1.6979, 5.2161]		
Imbens–Manski 95% CI				[1.5035, 5.2091]			

Rows trimmed by quartile: 632 of 13,718; 56 of 22,490; 117 of 29,022; 263 of 18,106. The untrimmed point estimate here (3.4077) differs slightly from Table 5 (3.407) because this sample additionally requires a matched cell-based exposure. Observations for the bound regressions: [TODO: confirm]. Outcome regressions use the occupation-level index and cluster by occupation.

monotonic age profile in the hours penalty itself, independent of exposure (Mother $\times$ age group 4: -2.083^{***} ; group 6: 1.063^* ; group 7: 1.337^*).

Table 8: Hours DDD: sensitivity of Mother $\times$ Post $\times$ WFH_Exposure

Specification	Coefficient	SE	N
<i>Exposure measure</i>			
Calibrated (primary)	3.407***	0.9302	275,708
External (Dingel–Neiman)	0.9404	0.8649	275,708
Realized (2021 anchor)	6.416*	2.454	275,390
<i>Age-balance</i>			
Age-interacted (+ Mother $\times$ age group)	3.367**	0.9553	275,708
Rewighted (pre-period age-group raking)	2.836**	0.9055	271,621

All rows: weekly hours, employed women, standard controls, clustered by occupation.

5.4 Heterogeneity by population group and the gender placebo

Table 9 re-estimates the hours DiD and DDD separately for Jewish and Arab women, and on men as a placebo. For Jewish women the pattern is the pooled pattern: the DiD is 0.9029^{***} (SE 0.2047) and the DDD 2.9238^{**} (SE 0.8901), each with a confidence interval that excludes zero. For Arab women the DiD is 0.1267 (SE 0.4798), indistinguishable from zero — there is no significant average hours effect for Arab mothers to begin with, and Arab women’s hours fell in the post-period overall (Post = -1.579^{***}) — while the

DDD is 6.0062* (SE 2.6426, 95% CI [0.8267, 11.1857]), nominally significant and twice the Jewish point estimate.

We do not read the larger Arab estimate as evidence that the mechanism is stronger for Arab women, for two reasons. First, the two coefficients are not statistically distinguishable. The Arab subsample is about nine times smaller than the Jewish one (26,164 versus 236,554 employed, occupation-matched women), the Arab standard error is nearly three times the Jewish one, and a two-sample z -test on the DDD term gives $z = (6.0062 - 2.9238) / \sqrt{2.6426^2 + 0.8901^2} = 1.105$, $p = 0.269$ — the apparent gap is well within what the Arab subsample’s own sampling noise can produce. (The same test on the DiD gives $z = 1.488$, $p = 0.137$.) Second, there is a plausible mechanical cause for the imprecision: Arab women’s employment in Israel is concentrated in a narrow set of occupations, disproportionately education and health-aide roles, so that comparatively few two-digit occupations carry enough Arab respondents to inform an occupation-specific comparison. With few occupations carrying real information, a small number of high-leverage occupations can dominate the estimate. The Arab result is therefore best described as imprecisely estimated and statistically indistinguishable from the Jewish estimate — a small-sample, thin-occupational-coverage fragility rather than a documented heterogeneous treatment effect. The Jewish-women estimate is the more credible of the two and anchors any subgroup claim we make.

The gender placebo goes the other way. Re-estimating the same models on men, with “mother” read as “father,” gives a DiD of -0.3860^* (SE 0.1909, $N = 276,343$) — fathers’ hours moved slightly *down* relative to childless men after 2021 — and a DDD of -1.8806 (SE 1.1226, $N = 266,087$), insignificant and with a negative point estimate. The women’s and men’s coefficients are statistically distinguishable: $z = 3.627$ ($p = 0.0003$) for the DDD and $z = 4.584$ ($p < 0.0001$) for the DiD. This is a stronger statement than “the placebo is insignificant”: fathers show no positive WFH-linked hours response, and a macro trend common to all workers cannot explain a positive shift for mothers alongside a negative one for fathers.

6 Discussion

Among mothers who remain employed, the post-2021 hours effect is not uniform across occupations. It is concentrated in occupations with higher WFH exposure, with a triple-interaction coefficient of about 3.4 additional weekly hours per unit of exposure, and it survives a correction for selection into employment. This is consistent with WFH access relaxing a binding time constraint specifically for mothers in teleworkable jobs — the Goldin (2014) mechanism — rather than with a general post-pandemic recovery common to all women, since the effect scales with occupational exposure rather than appearing uniformly, and since fathers show no analogous response.

Table 9: Hours DiD and DDD by subgroup, and the gender placebo

Subgroup	DiD: Mother $\times$ Post			DDD: Mother $\times$ Post $\times$ WFH_Exposure		
	Coef.	SE	N	Coef.	SE	N
All women (primary)	0.8261***	0.1830	281,750	3.4073***	0.9302	275,708
Jewish women	0.9029***	0.2047	241,738	2.9238**	0.8901	236,554
Arab women	0.1267	0.4798	26,876	6.0062*	2.6426	26,164
Men (placebo)	-0.3860*	0.1909	276,343	-1.8806	1.1226	266,087
95% CI, Jewish	[0.5017, 1.3041]			[1.1792, 4.6683]		
95% CI, Arab	[-0.8137, 1.0671]			[0.8267, 11.1857]		
Jewish vs. Arab, z (p)	1.488 (0.137)			1.105 (0.269)		
Women vs. men, z (p)	4.584 (< 0.0001)			3.627 (0.0003)		

Weekly hours, employed only, standard controls. DiD clustered by individual; DDD clustered by occupation. Men: “Mother” read as “Father.” Two-sample z -tests are valid without a covariance correction because the compared samples are disjoint.

Which occupations are doing the work is concrete. The calibration step moved ten occupations away from their US task-based scores toward realized Israeli behavior: clerical support and teaching were corrected sharply downward, because Israeli institutional practice kept them in person regardless of task content, while ICT professionals remained high on both measures. The triple interaction is identified off exactly this contrast, which grounds “exposure” in checkable occupational content. That the raw external index yields a positive but insignificant estimate is itself informative: the mechanism shows up cleanly only once exposure is measured as Israeli jobs were actually done, not as US task content says they could be.

The timing is also informative. The event study rises gradually from an insignificant 2021 estimate to a significant one in 2023, consistent with WFH adoption and household time reallocation being a multi-year adjustment rather than an instantaneous response to the 2020–21 shock — plausible given that occupational WFH norms were themselves still settling over 2021–2023.

Why hours and not employment? The employment DDD is not a failed test of the mechanism; it is an uninformative one, with a minimum detectable effect of a quarter of the baseline employment rate. The intensive-margin design is well powered because it can use occupation-level exposure, which the employment design cannot without conditioning its own outcome on itself. Substantively, the two results are coherent: the effect operates on *how much* mothers who stay employed work, not on *whether* they are employed, which is precisely the margin [Kleven et al. \(2019b\)](#) identify as a leading channel of the child penalty. This is also where our findings depart from [Harrington and Kahn \(2025\)](#): their US evidence is on employment and income, and we cannot confirm or reject an employment effect in Israel with this design; what we can say is that the intensive margin responds, and that it responds through remote work. A purely statistical account is not

mutually exclusive: the extensive-margin DDD requires an implausibly large true effect (on the order of a quarter of the baseline employment rate) to be detectable at all. The continuous hours outcome, by contrast, carries strictly more information per observation and is correspondingly better powered to detect an effect of realistic magnitude. One institutional account is also consistent with this pattern: Israel’s furlough (*Halat*) system preserves the formal employment relationship through disruption at comparatively low cost to the worker. Labor-supply adjustment is thus likely to occur first, or exclusively, on the less costly margin — hours — before reaching the more costly margin of exiting employment altogether. If this interpretation is correct, it carries a direct policy implication: expanding WFH availability should be evaluated as a lever for improving the quality and intensity of employment among mothers already in the labor force, rather than as an instrument expected to raise labor-force participation.

The subgroup results are more modest than the point estimates suggest. The Jewish-women estimate carries the pooled result. The Arab-women estimate is larger but imprecise, statistically indistinguishable from the Jewish one, and rests on thin occupational coverage; we report it transparently but do not claim heterogeneity on its basis.

7 Limitations

Parallel trends and the 2017 anomaly. The hours event study shows a statistically significant pre-trend violation in 2017 (-1.524^{***} relative to 2019), and a joint test on 2017–2018 rejects parallel trends over the full pre-period ($F(2, 65,088) = 23.7$, $p = 5.2 \times 10^{-11}$). By contrast 2018 is flat relative to 2019. The paper’s parallel-trends assumption therefore rests on the 2018–2019 comparison, not the full 2017–2019 window, and the 2017 anomaly is a limitation rather than something to be smoothed over. The problem is specific to hours: the same test on employment passes ($F(2, 79,069) = 0.63$, $p = 0.53$). The Lee-bounds counterfactual itself relies on the same parallel-trends assumption already dented by the 2017 finding.

Lee bounds: direction and monotonicity. The trimming construction only handles *excess* selection in the post-period mother cell — the direction the project’s own hypothesis predicts. Under-selection in that cell (fewer employed mothers than the counterfactual implies) is a missing-data problem, not an excess-observed-data problem, and is not addressed. The construction assumes monotone selection (that WFH weakly increases, never decreases, a mother’s employment probability) to justify trimming from a single tail, and computes selection rates unweighted, consistent with the rest of the analysis. A finding of small excess selection should be read as “this construction found little to correct,” not as proof that the selection concern is absent.

Survey weights. CBS survey weights are intentionally not applied in any outcome regression. This is a scope decision, not an oversight: reported rates and coefficients are unweighted estimates on the analysis sample, not population-representative statistics. The one exception is internal to constructing the cell-based exposure index, where occupation exposure is aggregated to demographic cells using the survey weight; that is not a representativeness correction for the outcome regressions.

The exposure anchor: 2021, not 2020. The realized WFH index is anchored to 2021 rather than to the 2020 variable the original research plan specified, because no 2020 extract exists. The justification is that 2021 already passes the validated data pipeline and is arguably a more stable measurement point than 2020’s emergency disruption. The downside is real: by 2021 occupations had had a year to adapt, and if adaptation was uneven a 2021-based index could rank occupations’ exposure differently than a 2020-based one would have. The realized index enters only as a robustness variant; the primary calibrated index draws on 2022–23 realized data for its swap test and is itself a documented compromise rather than a fully pre-treatment measure.

Age controls and age imbalance. Age enters as a categorical age-group code, because the extract has no continuous age or birth-year variable. Mothers and non-mothers are imbalanced on this code in the pre-period, and the imbalance tracks exposure quartile; the age-interacted and reweighted specifications address this and the estimate survives, but the reweighted attenuation ($3.407 \rightarrow 2.836$) is the largest single move any robustness check produces.

Inference. The hours DDD clusters on about 40 occupation groups and the employment DDD on about 210 demographic cells; no small-cluster correction is applied. The same person can in principle contribute observations to both the pre- and post-period. Occupation codes masked by the CBS for disclosure control are dropped when building exposure; a proxy check finds no strong evidence that this biases the exposure measure, though that is weak rather than conclusive evidence. The calibrated exposure measure uses post-period (2022–23) realized data, as noted.

8 Conclusion

[DRAFT CONCLUSION – FOR SELF-COMPARISON ONLY, NOT FOR SUBMISSION]

This paper asked whether the post-COVID shift to remote work altered the motherhood penalty in Israel, extending the question Harrington and Kahn

(2025) pose for the United States to a different country and to the intensive margin of labor supply. Using CBS Labor Force Survey microdata on women aged 25–59 across 2017–2019 and 2021–2023, we find no change in the mother/non-mother employment gap and an employment-by-exposure triple difference that is too underpowered to speak to the mechanism either way. On hours, by contrast, employed mothers gained about 0.83 weekly hours relative to childless women, and that gain is entirely concentrated in occupations exposed to remote work: the triple interaction of 3.41 hours per unit of exposure is significant at the 0.1% level, robust to a generalized Lee-bounds correction for selection into employment (Imbens–Manski 95% CI [1.50, 5.21]), to age-balance adjustments, and to a gender placebo in which fathers show, if anything, the opposite pattern. The result is consistent with remote work relaxing the temporal inflexibility that Goldin identifies as the last barrier to gender convergence, operating for mothers who remain employed in jobs that can actually be done from home. Its limitations are equally clear: a 2017 pre-trend anomaly confines the parallel-trends claim to 2018–2019, the selection correction handles only excess selection, the exposure measure is anchored a year later than planned and partly on post-period data, and the extensive margin remains, for this design, an open question rather than a null. The natural next steps are a higher-powered test of the employment margin and a jointly estimated model that lets the mechanism differ by sex and population group within a single specification.

[END DRAFT CONCLUSION]

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